

Question	Answer	Marks
8(a)	quantum of energy	M1
	of electromagnetic radiation	A1
8(b)(i)	$(\Delta)E = hc / \lambda$	C1
	$\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (1.96 \times 1.60 \times 10^{-19})$	C1
	$= 6.3 \times 10^{-7} \text{m}$	A1
8(b)(ii)	number per unit time = power / energy per photon $= (1.0 \times 10^{-2}) / (1.96 \times 1.60 \times 10^{-19})$ $= 3.2 \times 10^{16} \text{s}^{-1}$	A1

Question	Answer	Marks
8(b)(iii)	<i>either:</i> force = rate of change of momentum <i>or:</i> $F = \Delta p / t$	C1
	$p = E / c$	C1
	half the photons have change in momentum p , the other half have change in momentum $2p$	C1
	$F = [(1.96 \times 1.60 \times 10^{-19}) / (3.00 \times 10^8)] \times 3.2 \times 10^{16} \times [(2 + 1) / 2]$ $= 5.0 \times 10^{-11} \text{ N}$	A1

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